Vanamala Aman Preetham

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.83
Sri Chaitanya Junior Kalasala, Bagh Amberpet, Hyderabad	Class XII (Board of Intermediate Education, Telangana)	2023	98.80%
Slate The School, Gunfoundry, Hyderabad	Class X (Board of Secondary Education, Telangana)	2021	10.00%

SCHOLASTIC ACHIEVEMENTS

- KVPY Scholar SA-stream 2021
- Secured International 13th in SOF IMO Level-1 in Class 9
- 1200+ max rating in Codeforces
- Completed Abacus 10 levels by Class 4

WORK EXPERIENCE

Microsoft - Software Engineer Intern:

- Architected and implemented 4 of planned 6 agents in a **multi-agent AI automation framework**, defining the agent decomposition, task boundaries, orchestration strategy, and structured contracts based on customer and workflow requirements.
- Developed Python-based orchestration around **Microsoft Foundry** agents, implementing structured contract handling, agent invocation, execution polling, workflow coordination, and integration across specialized agent capabilities.
- Developed resilient **Playwright MCP** browser-automation workflows, creating detailed agent instruction sets with state-aware recovery loops, retries, validation, failure handling, and navigation recovery to handle unexpected application states and execution failures.
- Deployed and integrated the orchestration stack across **Azure Function App** and **Azure Container Apps**, implementing agent invocation, contract-based communication, execution polling, and cloud-service integration while contributing to end-to-end testing, debugging, and technical demonstrations.

Teaching Assistant for Introduction to Programming Course 2024

Teaching Assistant for Computer Networks Course 2026

SKILLS

- Programming Languages: C, C++, Python, JavaScript, RISC-V Assembly
- Technologies, Tools: SQL, PostgreSQL, MongoDB, HTML, CSS, Node.js, Express.js, FastAPI, React, TypeScript, ROS 2, Git, Playwright MCP, Microsoft Foundry, Azure Function App, Azure Container Apps
- Participated in IICPC Summer camp 2024

PROJECTS

Airline Reservation System and Dynamic Flight Scheduling:

- Designed core logic for dynamic ticket pricing, multi-routing, and dynamic flight scheduling using relational modelling, triggers, and transaction-safe queries in PostgreSQL.
- Collaborated on a locally hosted booking platform with integrated features like seat map-based selection, auto-updating fare display, booking confirmation flow and categorized ticket history.
- Drove system-level planning and database design, aligning schema decisions with application needs and coordinating feature integration across modules.

Infinite Precision Arithmetic for Integer and Floating Point Numbers:

- Implemented addition, subtraction, multiplication and division on arbitrary-precision integers and floating-point numbers using object oriented design in C++ Integrated modular design with
- Makefile for build automation and utilised
- Git for version control for collaborative development Achieved full correctness by passing all edge-case test cases

Emotion-Cause Identification and Summarization in Conversations:

- Implemented emotion-cause extraction by constructing emotion-candidate utterance pairs and training a contextual binary classifier to identify causal relationships in multi-turn dialogues.
- Enhanced abstractive summarization with emotion-aware prompting, conditioning summary generation on predicted emotion labels alongside the dialogue to incorporate emotional context.
- Integrated and evaluated the developed components within a multi-stage NLP pipeline spanning emotion classification, cause extraction, and abstractive summarization using transformer-based models.

FPT algorithm for the Minimum 2-Interval Hitting Set Problem:

- Analyzed the Minimum 2-Interval Hitting Set problem through an individual research study, investigating parameterized algorithms, structural reductions, and improved kernelization.
- Derived structural characterizations by analyzing solution configurations, identifying invariants and propositions, and restricting the search space to candidate points derived from a minimum 1-Interval Hitting Set.
- Designed an FPT-style bounded-search framework over the restricted candidate set, incorporating coverage tracking, feasibility checks, jump pointers, and maximum-matching-based lower-bound pruning to reduce the search space.

GraphX - High-Performance Graph Domain-Specific Language:

- Architected GraphX from its inception, owning the DSL specification and design around graph algebra and density-aware optimization; authored the language white paper and evaluated features for feasibility and coherence.
- Implemented the lexer and parser from the language specification, while debugging semantic-analysis and code-generation issues across the compiler pipeline.
- Designed the IGraph abstraction and Graph Factory, enabling density-aware selection among adjacency-list, adjacency-matrix, and hybrid backends for transpilation to optimized C++17.
- Validated the implementation through white-box testing, developing additional test scenarios beyond the team's primary cases and feeding findings back into implementation.

RELEVANT COURSES

- **Computer Science:** Data Structures, Algorithms, Software Engineering, Computer Networks, Database Management Systems, Compilers, Operating Systems, Computer Architecture, Theory of Computation, Introduction to Programming
- **Mathematics:** Linear Algebra, Probability and Random Variables, Introduction to Group Theory, Introduction to Statistics, Transform Techniques, Calculus, Vector Calculus, Discrete Maths for Computer Science
- **AI:** Foundations of Machine Learning, Deep Learning, Natural Language Processing, Linear Optimization, Convex Optimization, Introduction to AI
- **Theoretical Computer Science:** Computational Number Theory and Algebra, Algorithmic Techniques for Large Graphs, Theory of Computation, Approximation Algorithms
- **Entrepreneurship:** International Business, Innovation Management, Introduction to Entrepreneurship
- **Miscellaneous:** Communication Skills, Ethics and Values, Digital Fabrication

POSITION OF RESPONSIBILITY

- Core member of Vibes Club

EXTRACURRICULAR

- **Black Belt - Dan-1 in Karate**, with **3 world records** for **200+ kicks in under 2 minutes**
- **SGF National-Level Participant** in both **Karate** and **Tang Soo Do**, selected through competitive state-level trials

- **Grade 8 Certification in Plectrum Guitar** with **Merit** from **Trinity College London** -- the highest performance certification
- **Winner - 146th Brilliant Chess Trophy**, with **over 60 medals** across Karate, Olympiads, and Chess